



AI Circuit Architect

From a plain-English idea to a structured KiCad starting point — designed by a society of agents

Qwen Hackathon · Track: Agent Society

A schematic is slow to start — and one LLM cuts corners

The manual way

- **Blank-page problem**

Turning an idea into a structured KiCad project takes an experienced engineer.

- **Hours before the real work**

Sheet hierarchy, power domains, interfaces — all by hand.

- **Easy to forget the boring bits**

Protection, debug access, review items slip through under time pressure.

A single LLM

- **Fast, but shallow**

One pass gives a plausible answer — and skips the unglamorous essentials.

- **Misses safety / review items**

Reverse-polarity, overcurrent, reset/clock — gone in 1 of every 2 designs.

- **No one checks its work**

Nothing challenges the first draft or asks what's missing.

Our take: a small **team of agents that argue** → a structured, reviewed starting point you finish in KiCad.

Two heads beat one — and we measured it

+3.2

more design concerns surfaced than a single LLM — on average, across 5 diverse hard designs (live Qwen).

Multi-agent team



Single agent



Coverage = engineering concern surfaced as work, scored by a deterministic 12-point rubric.

Per design · coverage delta (multi – single)

Motor + safety

+2

Precision analog

+2

Battery IoT

+4

Medical wearable

+5

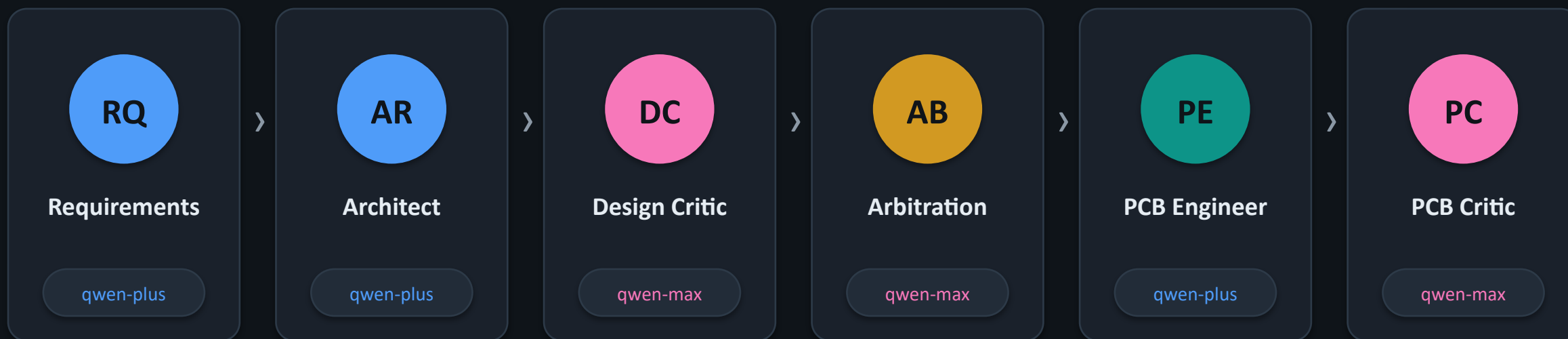
Industrial gateway

+3

↑ the gap widens with complexity

A team, not a prompt — six specialist agents

Each stage has one job and its own model tier. Work hands off left to right; reviewers can send it back.



Juniors (qwen-plus) propose; **senior reviewers (qwen-max) hold them to account — and demand rework.**

A stateless orchestrator coordinates them; you can run it one-click or sign off each stage.

When agents disagree, the design gets better

Our system is not a pipeline. The Critic challenges the Architect, who reworks — live, in rounds.



Live Agent-Society view · the Critic → Architect rework round

1

Round 1 — Design Critic

Flags a gap: the Architect's 5 blocks miss a Debug / SWD + status-LED block.

2

🔄 Rework — System Architect

Reacts to the critique and revises: adds the Debug block → 6 blocks.

3

Round 2 — Design Critic

Re-reviews the revision: no missing blocks remain. Conflict resolved.

This negotiation loop — not a single pass — is what raises coverage.

Measured on a deterministic rubric — not vibes

One 12-point engineering rubric scores both teams the same way — no LLM judging itself.

What a single agent most often skipped

across the 5 live designs — caught by the Critic / Arbitration

Reverse-polarity protection

5×

Overcurrent / fuse protection

4×

Docs & explicit uncertainty

3×

Surge / ESD on power input

2×

Clock source

2×

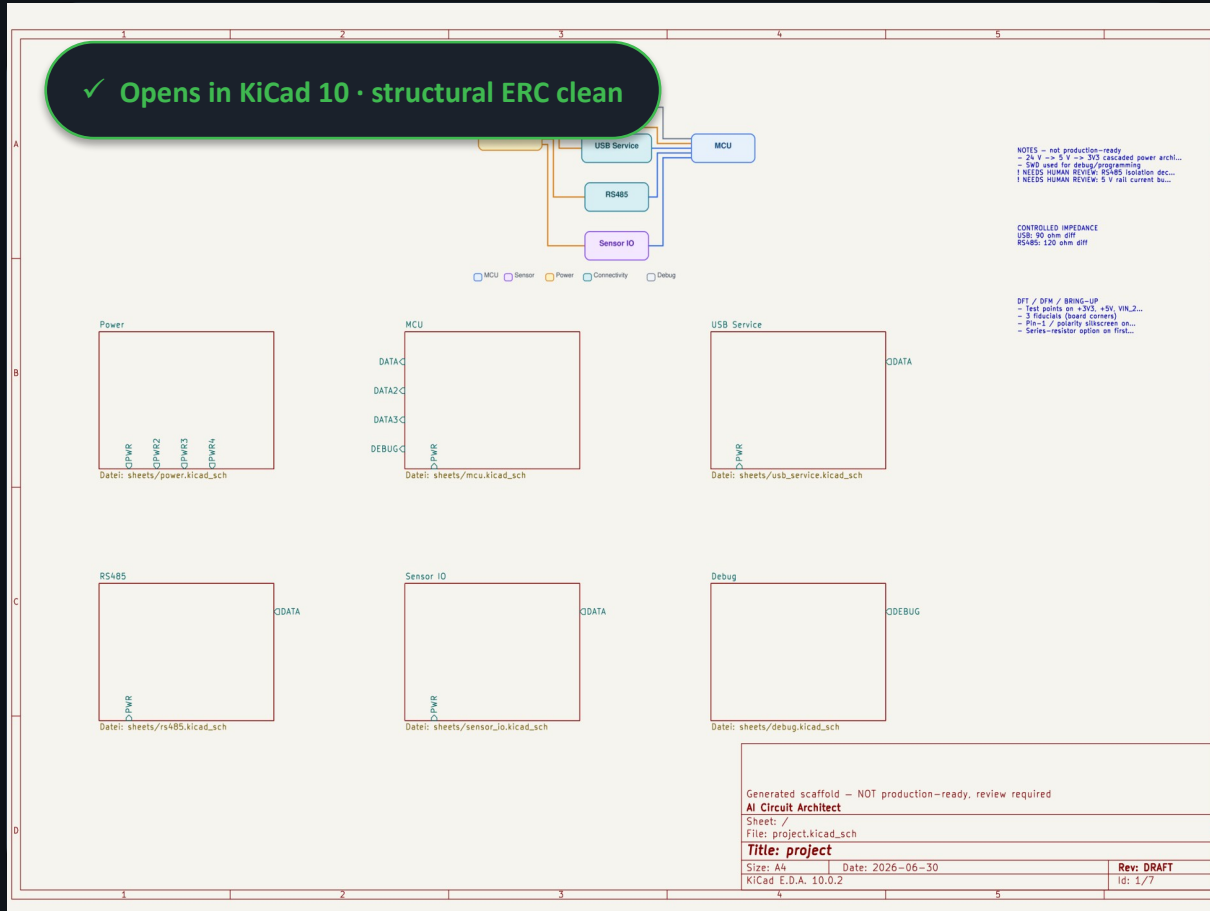
Average coverage

11.6 / 12 multi

8.4 / 12 single

The team's edge isn't cleverness — it's the review step that refuses to forget the unglamorous essentials.

What you actually get — a structured KiCad start



- **Hierarchical KiCad scaffold**

Sheets, power-port symbols, sheet pins and an embedded block diagram — opens & passes structural checks.

- **PCB-readiness pack (prep)**

Net classes, constraints (.kicad_dru), candidate parts, floorplan zones, a DFT/DFM checklist.

- **PDF report + KiCad ZIP**

A shareable report and the full project, downloadable in one click.

An engineering **scaffold, not a finished design** — placeholder blocks the engineer completes. No production claims; a human stays responsible.

AI prepares, the human decides

You stay in control

- **Approve, re-run or stop — per agent**

Every stage is shown before the next one runs.

- **One-click or step-by-step**

Auto-run to watch the team, or sign off each stage yourself.

- **Audience personas**

Professional / Student / Maker re-tones every explanation.

Built with

Qwen

qwen-plus / qwen-max, OpenAI-compatible API

FastAPI + Alpine.js

Python backend, live single-page UI (ELK diagrams)

KiCad-CLI 10

Real open / export / structural-ERC validation

WeasyPrint · Docker

PDF report; one-image deploy

Three jury criteria, one society of agents



Decomposition & roles

Six specialist agents, each with a job and a model tier.



Conflict resolution

A real Critic → Architect rework loop, resolved in rounds.



Measured efficiency

+3.2 coverage points over a single agent (live, 5 designs).

+3.2

more engineering concerns surfaced — the
case for a society of agents over a single
model.

From a plain-English idea to a structured KiCad start — reviewed by a society of agents.